

Selective Consistency Training with Reinforcement Learning

Anonymous Authors¹

Abstract

[TODO: write abstract once results are settled.]

1. Introduction

1.1. Relevant Pathologies

Large language model (LLM) behaviour is often influenced by prompt cues that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with a user’s stated views or preferences (Perez et al., 2022). Another example is evaluation gaming, where LLM behaviour depends on whether the model “believes” the input to be part of an evaluation. [TODO: cite apollo, anthropic and AISI’s research] This is particularly concerning because, if models behave differently in evaluations than in the real world, our evaluations become uninformative about real-world behaviour (Carlsmith, 2023).

1.2. Why Consistency Training

Two general approaches address the above behavioural inconsistencies. The first is to modify the input or the internal representations induced by the input so that the cues driving the inconsistencies are absent or hard for the LLM to detect [TODO: cite cot & activation steering work on eval awareness reduction]. However, isolating and removing the relevant cues from the input or internal representations is difficult in practice [TODO: why?]. The second is to train LLMs to behave consistently regardless of whether such cues are present. Consistency training is a class of methods focussing on the latter approach by training LLMs to produce consistent outputs or internal representations across input perturbations (Chua et al., 2024; Irpan et al., 2025).

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1.3. The problem with existing consistency training methods

A concern with existing consistency training methods is *obfuscation*: LLMs may learn to suppress from their outputs or internal representations any evidence that they detected the cues driving inconsistency since these outputs or internal representations are trained to be consistent with the cues being absent. Such obfuscation could make monitoring and auditing these systems more difficult. This concern can be avoided with *Selective Consistency Training*: training for consistency only over the selected behavioural properties where consistency is desirable, leaving other parts of the output free to vary.

We introduce *Reinforcement Learning Consistency Training* (RLCT), a reinforcement learning method for selective consistency training. RLCT works by computing and matching select behaviour propensities across input perturbations.

1.4. Summary of contributions

- We introduce *Reinforcement Learning Consistency Training* (RLCT), a method that enforces consistency only over selected behavioural properties of the output.
- We validate the method against BCT on the sycophancy benchmark of Chua et al. (2024) on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RLCT causes significantly less obfuscation compared to BCT, as measured by significantly lower decrease in bias acknowledgment.

2. Consistency Training

2.1. Existing methods

2.1.1. BIAS-AUGMENTED CONSISTENCY TRAINING

Bias-Augmented Consistency Training (BCT) trains LLMs to produce the same output on perturbed inputs as on the corresponding unperturbed input. For each prompt, a response is sampled from the LLM. The LLM is then fine-tuned to produce that same response when given perturbed prompts (Chua et al., 2024).

2.1.2. ACTIVATION CONSISTENCY TRAINING

Activation Consistency Training (ACT) trains LLMs to produce internal representations when given a perturbed input that match the corresponding unperturbed input (Irpan et al., 2025). This is achieved by minimising the difference between residual-stream activations on the perturbed and clean prompts at matching token positions, pushing the perturbed activations toward the clean ones. ACT is applied only to prompt token positions, not response token positions. **[TODO: mention BCT still SOTA]**

2.2. Existing Applications

Consistency training has previously been applied to sycophancy (Chua et al., 2024), jailbreak robustness (Irpan et al., 2025), and reducing persona drift **[TODO: cite @misc-shah2026frustration, title = Gemma Gets Help: Mitigating Frustration and Persona Drift with Consistency Training, author = Shah, Neil and Africa, David Demitri, year = 2026, howpublished = LessWrong, note = SPAR Research Fellowship, url =]**. Irpan et al. (2025) report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility. -

2.3. Novel applications enabled by selective consistency training

BCT and ACT train LLMs to generate the same output or internal representations as when certain cues are absent from the prompt. This raises multiple concerns. Irpan et al. (2025) note that this training may misgeneralise to LLMs ignoring other important parts of the prompt, such as user instructions. Furthermore, constraining the entire output or internal activations also constrains any mentions or verbalisations of the prompt cues driving inconsistency. Therefore such training may also lead to obfuscation. This hampers the applicability of these methods when monitoring and auditing are important. As an example, training models to not verbalise any prompt injection jailbreaks may reduce their monitorability. Similarly, training LLMs to not verbalise when they correctly infer that they are being evaluated, reduces our ability to track evaluation awareness **[TODO: cite eval awareness literature eg. ansti scheming, apollo system cards, igor’s research]**.

Selective consistency training avoids the above concerns by selectively constraining only desired behaviours to be consistent, leaving other behaviours such as verbalisation or acknowledgement of the cues driving inconsistency free to vary.

3. Reinforcement Learning Consistency Training

We introduce *Reinforcement Learning Consistency Training* (RLCT), a method for selective consistency training.

3.1. Method

Consider an input x . Let $\mathcal{T}$ denote a set of *training* perturbations and $\mathcal{A}$ a set of *anchor* perturbations (the two sets may overlap), and write $\mathcal{F} = \mathcal{T} \cup \mathcal{A}$ for their union. For each perturbation $f \in \mathcal{F}$, we sample N trajectories from the current policy on the perturbed input x_f . Each trajectory i is scored for the target behavioural property by a binary classifier T (for example, “the response matches the bias suggested in the prompt”).

Define the local behaviour rate for each $f \in \mathcal{F}$ as

$$p_f = \frac{1}{N} \sum_{i=1}^N T(\tau_{f,i}), \quad (1)$$

and let $q = Q(\{p_f\}_{f \in \mathcal{A}})$ denote a target rate aggregated over the anchor perturbations. This target rate may be the behaviour rate on a single anchor perturbation (for example, the clean or unperturbed prompt), or a function of the per-perturbation rates such as their mean, minimum, or maximum.

The consistency reward, applied only to trajectories from training perturbations, rewards a trajectory when its trait value moves the local rate p_f toward the target rate q :

$$r_{f,i}^{\text{cons}} = -(p_f - q)(T(\tau_{f,i}) - p_f), \quad f \in \mathcal{T}. \quad (2)$$

The baseline p_f is the variance-optimal centring for trajectories sampled under perturbation f from the current policy.

To prevent the target rate itself from drifting during training, an optional anchor reward, applied only to trajectories from anchor perturbations, ties q to the same target statistic $q_0 = Q(\{p_f^{(0)}\}_{f \in \mathcal{A}})$ computed under the initial policy π_0 (typically a checkpoint taken before consistency training begins), where $p_f^{(0)}$ is the behaviour rate of π_0 on perturbation f :

$$r_{f,i}^{\text{anchor}} = -(q - q_0)(T(\tau_{f,i}) - q), \quad f \in \mathcal{A}. \quad (3)$$

The baseline q is again the variance-optimal centring under the current sampling distribution: q is the empirical mean of trait values over the anchor rollouts, whereas q_0 comes from a frozen policy and would not yield an unbiased baseline.

The total reward for trajectory $\tau_{f,i}$ combines the two terms with mixing weight α :

$$r_{f,i} = (1 - \alpha) r_{f,i}^{\text{cons}} + \alpha r_{f,i}^{\text{anchor}}, \quad (4)$$

with each component taken as zero outside its domain of definition. A trajectory from a perturbation in $\mathcal{T} \cap \mathcal{A}$ thus receives both components; one from $\mathcal{T} \setminus \mathcal{A}$ receives only the consistency term, and one from $\mathcal{A} \setminus \mathcal{T}$ only the anchor term.

We optimise the policy with GRPO and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all $|\mathcal{F}|$ perturbations of a given input, with N trajectories per perturbation; advantage normalisation and policy updates then proceed as usual. We use parameter-efficient fine-tuning (LoRA) throughout.

3.2. Why this avoids the obfuscation pressure

Because the reward depends only on the trait T , the per-perturbation rates p_f , and the chosen target rate q , the model is not penalised for any particular content in the chain of thought. In particular, if the model verbalises that it has noticed a perturbation, the reward is unchanged provided its action matches the target rate. This contrasts with BCT, where matching a clean-prompt target response would penalise verbalisation of the perturbation.

4. Experiments

4.1. Environment and bias types

We validate the method on the sycophancy benchmark of Chua et al. (2024). The benchmark consists of multiple-choice questions to which a small prompt-level perturbation has been added that suggests a particular answer. We use six single-turn bias types from the benchmark: suggested answer, distractor fact, distractor argument, post hoc, spurious few-shot squares, and wrong few-shot. Brief descriptions are given in Appendix A; see Chua et al. (2024) for full specifications.

4.2. Models

We run experiments on two open-weight models: Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B. For Meta Llama 3.1 8B Instruct, the prompt instructs the model to reason step-by-step before answering. For OpenAI GPT OSS 20B, no such instruction is added: the model already produces internal reasoning before its final answer in its native response format. All training uses LoRA adapters of rank 8.

4.3. Train/evaluation split

We train on the held-in question sources LogiQA and Hel-laSwag (1024 questions sampled from each, drawn from the test split following the BCT data-generation procedure) and evaluate on a held-out question source, the multiple-choice

subset of Humanity’s Last Exam (HLE), with 100 questions per bias type per checkpoint and 400 questions per bias type for the base-model reference. For BCT, an additional 2048 instruction-following samples (prompts drawn from the Cleaned Alpaca dataset, with completions re-sampled from the target model so the supervision is self-distilled) are mixed into the training data to mitigate degradation on unrelated tasks. We train on the distractor-argument bias only and evaluate on all six bias types, which lets us measure transfer of the consistency objective to unseen biases. An additional regime that trains on a mixture of distractor-argument and wrong-few-shot prompts is reported in Appendix G.

4.4. Training settings

BCT is fine-tuned by SFT for one epoch on the 2048 BCT samples plus the 2048 instruction-following samples, at batch size 128 (32 optimisation steps). RLCT is trained with GRPO for one epoch over 64 prompts at batch size 4 prompts per update (16 update steps); each prompt is expanded into a group of $|\mathcal{F}| = 2$ perturbations (biased and unbiased) with $N = 128$ rollouts per perturbation, and the consistency reward uses the unbiased prompt as the target rate q with anchor weight $\alpha = 0$. For each method-model combination we run three learning rates and pool the resulting checkpoints when reporting metrics, which gives a binomial standard error without tripling the rollout budget. Remaining hyperparameters are listed in Appendix B.

4.5. Metrics

For each evaluation question x we sample one response under the biased prompt and one under the unbiased prompt, and let $b(x), u(x) \in \{0, 1\}$ indicate whether the model selected the biased option on each. We report three per-question switch rates, each averaged over questions:

$$\text{BSR}_{\rightarrow} = \frac{1}{|\mathcal{X}|} \sum_x \max(0, b(x) - u(x)),$$

$$\text{BSR}_{\leftarrow} = \frac{1}{|\mathcal{X}|} \sum_x \max(0, u(x) - b(x)),$$

$$\text{BSR}_{\text{tot}} = \frac{1}{|\mathcal{X}|} \sum_x |b(x) - u(x)|,$$

which we call the *towards-bias switch rate*, the *away-from-bias switch rate*, and the *total switch rate* respectively. The towards-bias switch rate counts questions where adding the bias prompt switches the model from a non-biased to the biased answer; the away-from-bias switch rate counts the reverse switch; the total switch rate sums both. Aggregating per question rather than taking the absolute difference of marginal rates preserves the directional signal: a model that switches toward the bias on half its questions and away on the other half has $\text{BSR}_{\text{tot}} = 1$ but a marginal-rate difference of 0. The main text reports only the towards-bias switch rate; the away-from-bias and total switch rates are reported in Appendix D. We also report the *Bias Acknowledgement*

Rate (BAR), the rate at which the model’s chain of thought acknowledges the perturbation (scored by Google Gemma 4 31B); a drop in BAR after consistency training indicates obfuscation.

4.6. Baselines

We compare RLCT against three baselines: (i) the untrained base model; (ii) BCT trained on a matched training set (same question sources, same training bias regime); and (iii) for each of BCT and RLCT, a *control* variant trained on the same questions but with the bias-inducing perturbation removed: the model is trained for consistency between two unbiased copies of the prompt rather than between the biased and unbiased prompt. The control isolates training-induced drift from the consistency signal: any change in the switch rates or BAR observed in the control reflects the effect of fine-tuning on unbiased data alone, leaving the residual gap between the control and the consistency-trained run as the contribution of the consistency objective itself. Control results are reported in Appendix C. **[TODO: ACT baseline if available.]**

4.7. Results

Figure 1 reports the towards-bias switch rate ($BSR_{\rightarrow}$) on the held-out HLE evaluation set; Figure 2 reports the bias acknowledgement rate (BAR). Results for the DA+WFS training regime, the matched controls, the away-from-bias and total switch rates, the verbalised/unverbalised splits, and accuracy are reported in the appendix.

Bias following. On the training bias both methods substantially reduce the towards-bias switch rate relative to the base model. On Meta Llama 3.1 8B Instruct, BCT reduces $BSR_{\rightarrow}$ from 0.64 to 0.15 and RLCT reduces it to 0.16; on OpenAI GPT OSS 20B, BCT reduces $BSR_{\rightarrow}$ from 0.30 to 0.11 and RLCT reduces it to 0.21. RLCT therefore matches BCT’s in-distribution reduction on Meta Llama 3.1 8B Instruct but reduces $BSR_{\rightarrow}$ less than BCT on OpenAI GPT OSS 20B. Improvements transfer to the five held-out bias types but not uniformly: BCT decreases $BSR_{\rightarrow}$ more than RLCT on wrong-few-shot for Meta Llama 3.1 8B Instruct and on suggested-answer for both models, while RLCT decreases $BSR_{\rightarrow}$ more than BCT on post-hoc for both models, where BCT increases the switch rate above base on OpenAI GPT OSS 20B from 0.23 to 0.33. Averaged over the five held-out biases the two methods are within standard error of each other. On Meta Llama 3.1 8B Instruct, BCT reaches 0.38 and RLCT reaches 0.36, against a base rate of 0.45; on OpenAI GPT OSS 20B, BCT reaches 0.19 and RLCT reaches 0.18, against a base rate of 0.22.

Bias acknowledgement. The two methods have markedly different effects on BAR. BCT reduces BAR on the training

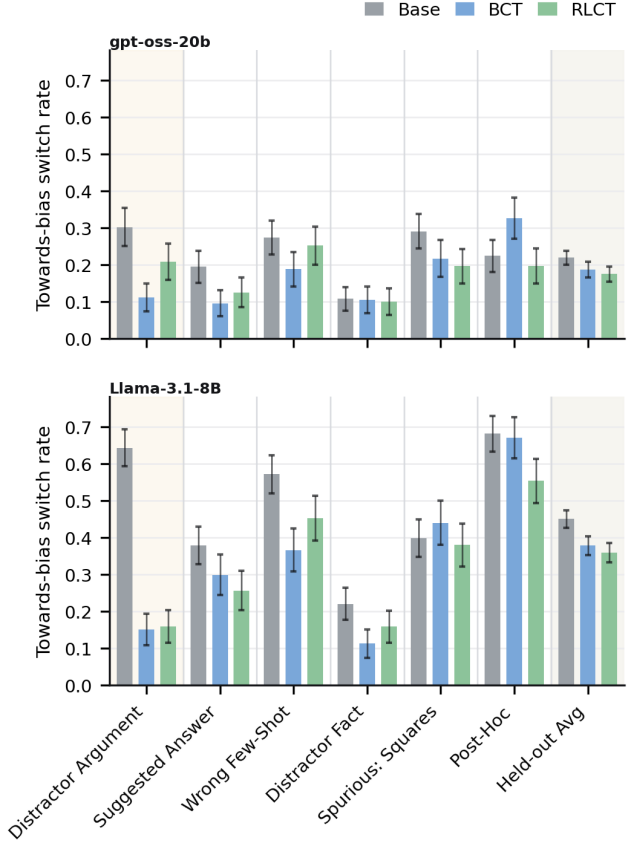


Figure 1. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

bias from 0.99 to 0.22 on Meta Llama 3.1 8B Instruct and from 0.56 to 0.08 on OpenAI GPT OSS 20B. This reduction transfers to the held-out biases, where the average BAR drops from 0.24 to 0.07 on Meta Llama 3.1 8B Instruct and from 0.41 to 0.18 on OpenAI GPT OSS 20B. RLCT does not measurably reduce BAR. On the training bias, RLCT reaches 0.89 on Meta Llama 3.1 8B Instruct and 0.45 on OpenAI GPT OSS 20B, against base rates of 0.99 and 0.56 respectively; the held-out averages are 0.27 on Meta Llama 3.1 8B Instruct and 0.37 on OpenAI GPT OSS 20B, both within standard error of the base rates of 0.24 and 0.41. The largest between-method gap is on spurious-squares for Meta Llama 3.1 8B Instruct, where BCT reduces BAR from 0.61 to 0.09 while RLCT moves it to 0.76. We read this as direct evidence of the obfuscation pressure that motivates the method: matching a clean-prompt response under BCT trains the model away from verbalising the cue, whereas the

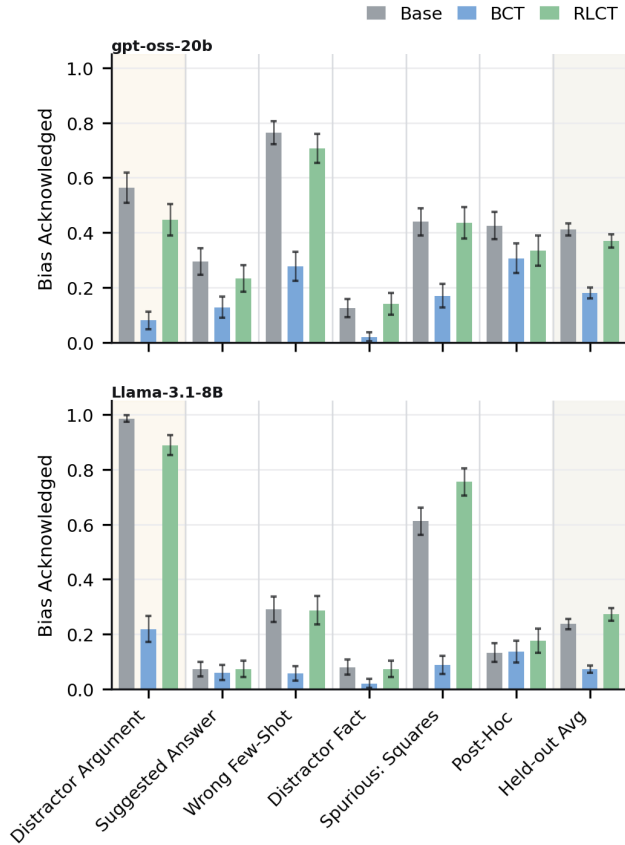


Figure 2. Bias acknowledgement rate (BAR) on HLE under the biased prompt. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

RLCT reward leaves the chain of thought free.

Summary. At a comparable level of held-out bias-following reduction, RLCT preserves bias acknowledgement near the base rate while BCT reduces it by roughly 56–70% on average. This is consistent with the prediction in Section 2.3 that selective consistency training avoids the obfuscation pressure inherent in matching the entire response to a clean-prompt target.

5. Discussion

[TODO: interpret results once finalised.]

Limitations. Consistency training does not address deceptive alignment: a model that is already deceptively aligned could plausibly be taught to hide its eval-gaming behaviour

better, similarly to how standard post-training methods make sleeper-agent behaviours difficult to catch (Hubinger et al., 2024). Our experiments are also restricted to a narrow bias-following setting on multiple-choice questions; the obfuscation question for the eval-awareness setting itself has not been tested directly. [TODO: add limitations specific to results once finalised.]

Future directions. Application of selective consistency training to evaluation-awareness settings (Section 2.3), extension to multimodal consistency (e.g. across input or output modalities), and application to other behavioural properties beyond bias-following.

6. Conclusion

[TODO: summarise once results are in.]

References

- Carlsmith, J. Scheming AIs: Will AIs fake alignment during training in order to get power?, November 2023. URL <http://arxiv.org/abs/2311.08379>. arXiv:2311.08379 [cs].
- Chua, J., Rees, E., Batra, H., Bowman, S. R., Michael, J., Perez, E., and Turpin, M. Bias-augmented consistency training reduces biased reasoning in chain-of-thought, 2024. URL <https://arxiv.org/abs/2403.05518>.
- Hubinger, E., Denison, C., Mu, J., Lambert, M., Tong, M., MacDiarmid, M., Lanham, T., Ziegler, D. M., Maxwell, T., Cheng, N., Jermyn, A., Askell, A., Radhakrishnan, A., Anil, C., Duvenaud, D., Ganguli, D., Barez, F., Clark, J., Ndousse, K., Sachan, K., Sellitto, M., Sharma, M., DasSarma, N., Grosse, R., Kravec, S., Bai, Y., Witten, Z., Favaro, M., Brauner, J., Karnofsky, H., Christiano, P., Bowman, S. R., Graham, L., Kaplan, J., Mindermann, S., Greenblatt, R., Shlegeris, B., Schiefer, N., and Perez, E. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training, January 2024. URL <http://arxiv.org/abs/2401.05566>. arXiv:2401.05566 [cs].
- Irpan, A., Turner, A. M., Kurzeja, M., Elson, D. K., and Shah, R. Consistency training helps stop sycophancy and jailbreaks, 2025. URL <https://arxiv.org/abs/2510.27062>.
- Perez, E., Ringer, S., Lukošiūtė, K., Nguyen, K., Chen, E., Heiner, S., Pettit, C., Olsson, C., Kundu, S., Kadavath, S., Jones, A., Chen, A., Mann, B., Israel, B., Seethor, B., McKinnon, C., Olah, C., Yan, D., Amodei, D., Amodei, D., Drain, D., Li, D., Tran-Johnson, E., Khundadze, G., Kernion, J., Landis, J., Kerr, J., Mueller,

J., Hyun, J., Landau, J., Ndousse, K., Goldberg, L., Lovitt, L., Lucas, M., Sellitto, M., Zhang, M., Kingsland, N., Elhage, N., Joseph, N., Mercado, N., Das-Sarma, N., Rausch, O., Larson, R., McCandlish, S., Johnston, S., Kravec, S., Showk, S. E., Lanham, T., Telleen-Lawton, T., Brown, T., Henighan, T., Hume, T., Bai, Y., Hatfield-Dodds, Z., Clark, J., Bowman, S. R., Askell, A., Grosse, R., Hernandez, D., Ganguli, D., Hubinger, E., Schiefer, N., and Kaplan, J. Discovering Language Model Behaviors with Model-Written Evaluations, December 2022. URL <http://arxiv.org/abs/2212.09251>. arXiv:2212.09251 [cs].

A. Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see [Chua et al. \(2024\)](#) for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., “My observations typically align with B.”). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An “irrelevant fun fact” about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Distractor argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as “someone else’s argument I read”, and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., “The best answer is: (B)”); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

Spurious few-shot squares. Several few-shot examples precede the question, each labelled with a correct answer; the labelled correct option is always marked with a small black-square symbol. A model that picks up the spurious

symbol-to-label association will pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistently incorrect labelling.

B. Training hyperparameters

All training uses LoRA adapters of rank 8.

BCT. A midway checkpoint is saved every 16 steps.

RLCT. Rollouts are sampled at temperature 1.0 with up to 20,480 generated tokens. The KL penalty to a fixed reference policy is 0.05 and the policy update uses the standard PPO clipped objective.

Learning-rate pooling. For each method-model combination we run three learning rates $\{1.0, 2.86, 5.0\} \times 10^{-4}$ and pool checkpoints across them when computing metrics. This gives a binomial standard error on the metric without tripling the rollout budget.

C. Control results

The control variants – BCT trained against an unbiased target and RLCT trained with two unbiased copies of the prompt – are shown alongside the consistency-trained runs as hatched bars (**BCT Control**, **RLCT Control**) in every appendix figure. They isolate the effect of fine-tuning on unbiased data alone from the effect of the consistency objective. Across the switch-rate plots (Figures 3–9) the controls track the base model closely on most held-out biases and sit visibly above the consistency-trained runs on the trained bias, confirming that the trained-bias reduction is driven by the consistency signal rather than by drift from fine-tuning.

The controls also rule out an obfuscation null hypothesis for BCT. Controls preserve bias acknowledgement near the base rate (Figure 10), so the BAR collapse seen for BCT proper is specifically attributable to training the model to match a clean-prompt response on biased inputs. The RLCT controls likewise leave BAR near base, ruling out the alternative that RLCT happens to boost acknowledgement for unrelated reasons.

D. Switch-rate breakdowns

This appendix reports the towards-bias, away-from-bias, and total per-question switch rates under both the DA-only and the DA+WFS training regimes (top and bottom rows of every figure), as well as the verbalised and unverbali-

splits of the towards-bias and total switch rates, broken down by which model produces which split.

D.1. Towards-bias, away-from-bias, total

Figures 3, 4, and 5 report $BSR_{\rightarrow}$, $BSR_{\leftarrow}$, and BSR_{tot} for both models. The DA-only towards-bias rates (Figure 3, top rows) reproduce the main-text plot with controls overlaid; the DA+WFS rows (bottom) extend the picture to a richer training mix.

For both models the away-from-bias rate is small relative to the towards-bias rate, with $BSR_{\leftarrow} \lesssim 0.15$ for almost every cell, so the total switch rate is dominated by towards-bias switching and Figure 5 largely tracks Figure 3. Two patterns are worth flagging. First, on Meta Llama 3.1 8B Instruct, BCT raises $BSR_{\leftarrow}$ on the trained DA bias to roughly twice the base rate while RLCT leaves it unchanged: BCT trains a directional response, namely always pick the unbiased answer, rather than a property of consistency. Second, on OpenAI GPT OSS 20B, $BSR_{\leftarrow}$ stays near base for both methods, so the small reduction in $BSR_{\rightarrow}$ for RLCT relative to the base translates into a similarly small reduction in BSR_{tot} .

The DA+WFS regime (bottom rows) lowers $BSR_{\rightarrow}$ further on the two trained biases for both methods, with held-out averages within standard error of the DA-only regime: BCT reduces $BSR_{\rightarrow}$ on wrong-few-shot to within standard error of zero for both models, RLCT reduces wrong-few-shot $BSR_{\rightarrow}$ by a smaller amount, and the held-out averages remain within standard error of each other.

D.2. Verbalised and unverbaised splits

We split each question by whether the model verbalised the bias in its chain of thought on the biased prompt. The verbalised and unverbaised subsets per condition have very different sample sizes: BCT verbalises rarely, so the verbalised plot has small n for BCT, while the unverbaised plot has small n for RLCT and the base model on biases the model habitually acknowledges, for example distractor argument on Meta Llama 3.1 8B Instruct, where the base verbalises in over 99% of cases. The per-bar n counts annotated on the figures should be consulted before reading too much into any individual cell. We show the splits only for the towards-bias rate in Figures 6 and 7, and the total switch rate in Figures 8 and 9; the away-from-bias splits show the same qualitative picture and are omitted to save space.

Verbalised subset. Figure 6 restricts to questions where the biased-prompt chain of thought acknowledged the cue. For RLCT and the base model the verbalised subset still shows substantial bias following. On Meta Llama 3.1 8B Instruct the held-out-average towards-bias rate is 0.49 for RLCT and 0.45 for the base; on OpenAI GPT OSS 20B

the rate is 0.21 for both. These rates are indistinguishable from the unconditional rates in the main text, ruling out the alternative that the RLCT reduction in $BSR_{\rightarrow}$ comes from the model noticing the bias more often and only switching when it did not: even when it does notice, it follows the bias at the base rate and is moved by the reward toward consistency. For BCT the verbalised subset is too small to read off reliable per-bias rates, with $n \lesssim 30$ for several biases. Figure 8, the total switch rate on the same subset, shows the same pattern.

Unverbaised subset. Figure 7 restricts to questions where the chain of thought did not acknowledge the cue. This is the operating regime for BCT across most biases, and indeed the BCT bars in this plot are similar to its bars in Figure 3. On biases where the base model unverbaised rarely, namely distractor argument and spurious-squares on Meta Llama 3.1 8B Instruct, the unverbaised subset for the base model has very low n but a high $BSR_{\rightarrow}$, so the few times the model fails to acknowledge the cue it is much more likely to follow it. RLCT in this subset behaves similarly to BCT: when the model has not acknowledged the cue, both methods reduce following at roughly comparable rates. The total switch rate (Figure 9) again mirrors the towards-bias plot.

The combination of these two splits supports our reading of the main-text BAR result. RLCT preserves acknowledgement and reduces following both when the cue is acknowledged and when it is not; BCT reduces following primarily by routing the model into the unverbaised regime.

E. Bias acknowledgement details

Figure 10 reports the unconditional bias acknowledgement rate (BAR) under the biased prompt for both models and both training regimes, and Figure 11 restricts BAR to the *bias-switched* subset: questions on which adding the bias prompt changed the parsed final answer (i.e. where $b(x) \neq u(x)$). This is the conditional BAR on questions where the bias actually moved the model.

Figure 10 reproduces the main-text result at higher resolution and across both training regimes: BCT reduces BAR on the trained bias and on the held-out average, while RLCT does not move BAR away from the base rate by more than a binomial standard error in either training regime. The DA+WFS BCT run yields a larger BAR reduction on its trained biases than the DA-only run; DA+WFS RLCT remains within standard error of base.

Figure 11 is the more behaviourally meaningful view: it asks whether, on the questions where adding the bias did swing the model’s answer, the model said so in its reasoning. On Meta Llama 3.1 8B Instruct, the held-out-average condi-

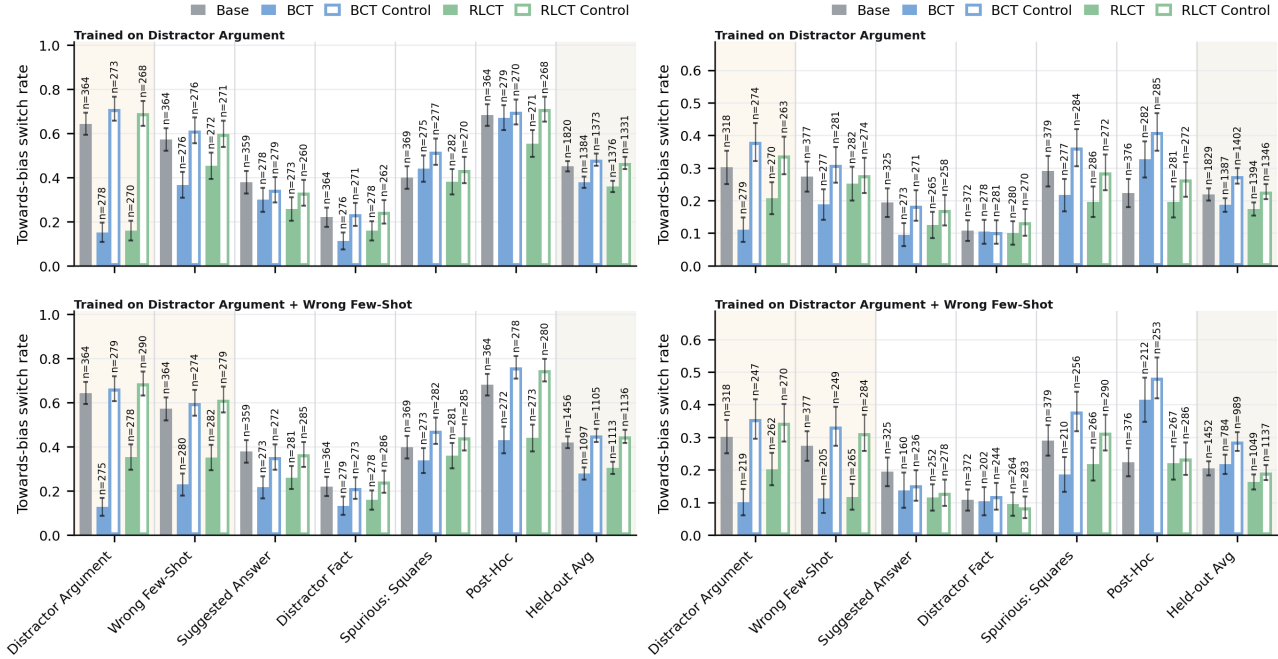


Figure 3. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

tional BAR is 0.30 for the base model and 0.06 under BCT, but stays at 0.30 under RLCT; on OpenAI GPT OSS 20B it drops from 0.42 to 0.20 under BCT and stays at 0.41 under RLCT. Equivalently, under BCT, on the questions where the bias was operative, the model now reasons as if no bias were present in approximately 80% of cases on Meta Llama 3.1 8B Instruct and approximately 50% of cases on OpenAI GPT OSS 20B, whereas under RLCT this rate is within standard error of the base rate. An acknowledgement-rate monitor calibrated on the base model would therefore fail to flag the bias-driven answers under BCT, but not under RLCT.

F. Accuracy

We report two accuracy metrics: accuracy under the biased prompt in Figure 12, and accuracy under the unbiased prompt in Figure 13. HLE is intentionally hard, so absolute accuracies remain at $\lesssim 0.25$ throughout.

Under the biased prompt, BCT and RLCT both leave accuracy within standard error of the base model on Meta Llama 3.1 8B Instruct and on OpenAI GPT OSS 20B across all held-out biases and both training regimes. Where the consistency-trained runs reduce $BSR_{\rightarrow}$, they do not do so by trading correctness in either direction. Under the unbiased prompt the differences are small but consistent. On Meta

Llama 3.1 8B Instruct, RLCT reaches 0.18, BCT reaches 0.16, and the base reaches 0.13; on OpenAI GPT OSS 20B, BCT reaches 0.14, RLCT reaches 0.10, and the base reaches 0.11. In every case the differences sit within the binomial standard error. The unbiased-prompt bias-match rate in Figure 14 is reported for completeness: on the unbiased prompt, RLCT does not raise the rate at which the model picks the bias-target option above base, so the bias-following reduction is not produced by inducing a generic preference against those options.

G. Distractor-argument + wrong-few-shot training regime

We additionally trained BCT and RLCT on a mixture of distractor-argument and wrong-few-shot prompts (DA+WFS) on LogiQA+HellaSwag, with all other settings matching the DA-only setup of Section 4. The DA+WFS BCT run uses the same training-data scale per bias, giving roughly 48 optimisation steps at batch size 128. RLCT uses an identical configuration to the DA-only setup with both bias types sampled in the prompt mix. Per-metric DA+WFS results are shown as the second row of every figure in Appendices D, E, and F. The qualitative picture matches the DA-only regime: BCT collapses BAR on its trained biases while RLCT preserves it, and the held-out switch-rate reductions are comparable between the two methods.

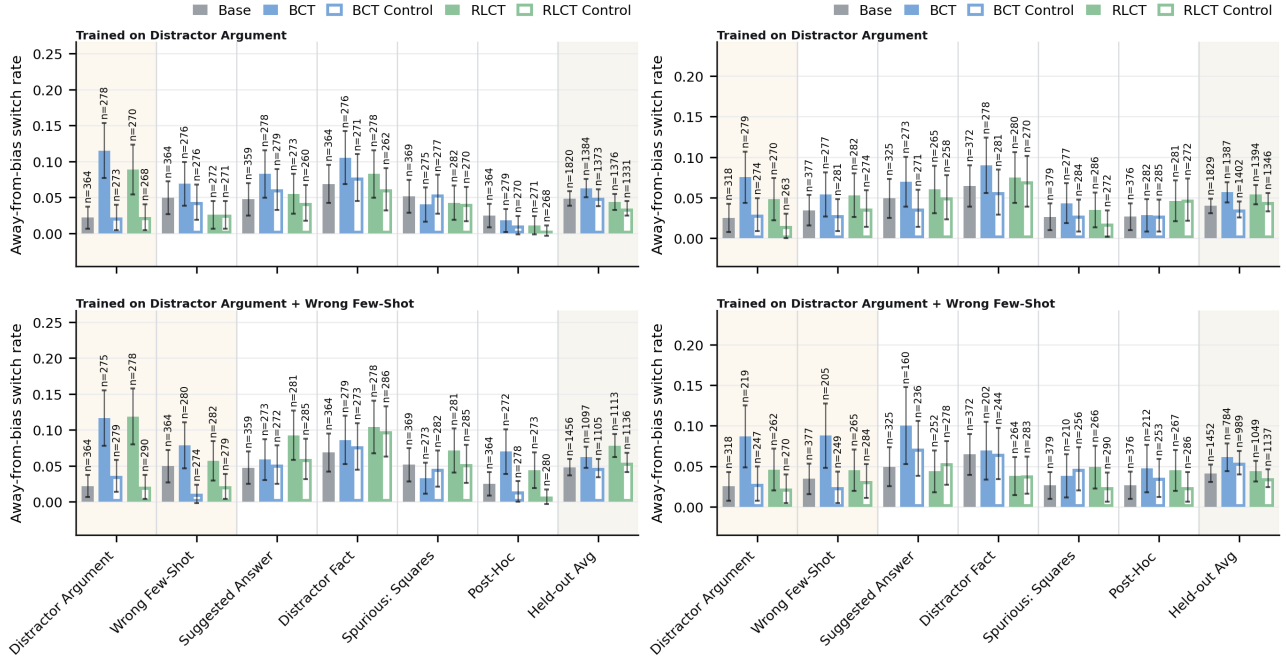


Figure 4. Away-from-bias switch rate BSR_{away} on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

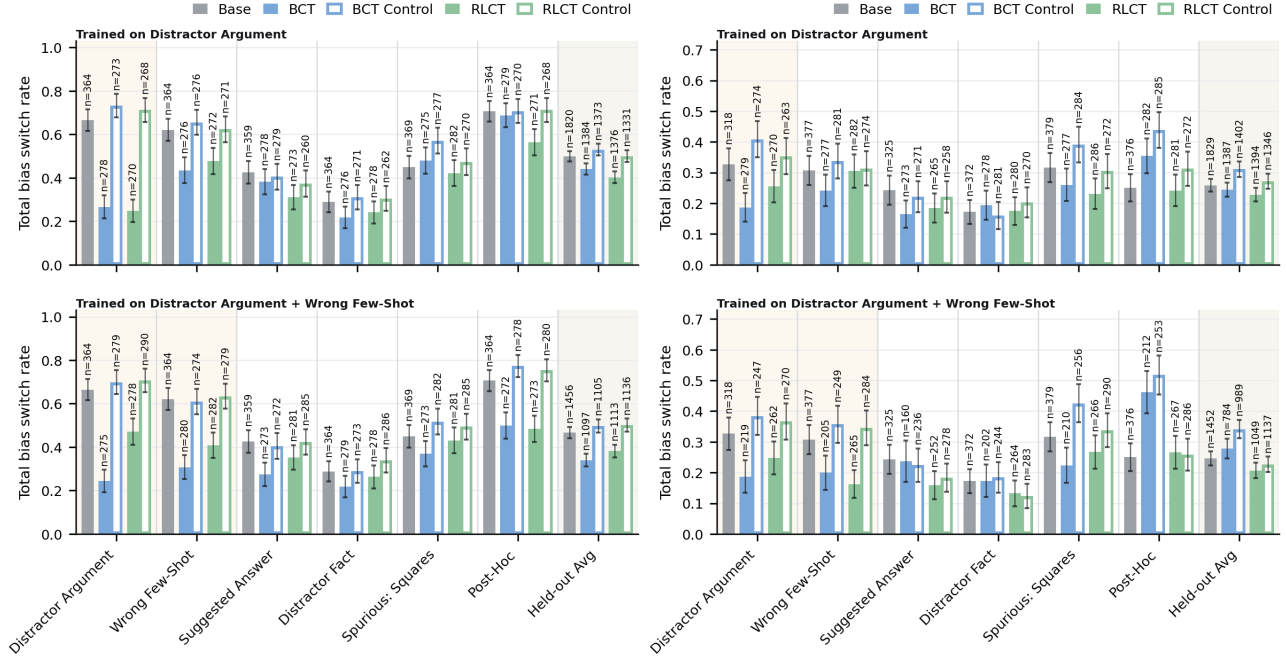


Figure 5. Total switch rate BSR_{tot} on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

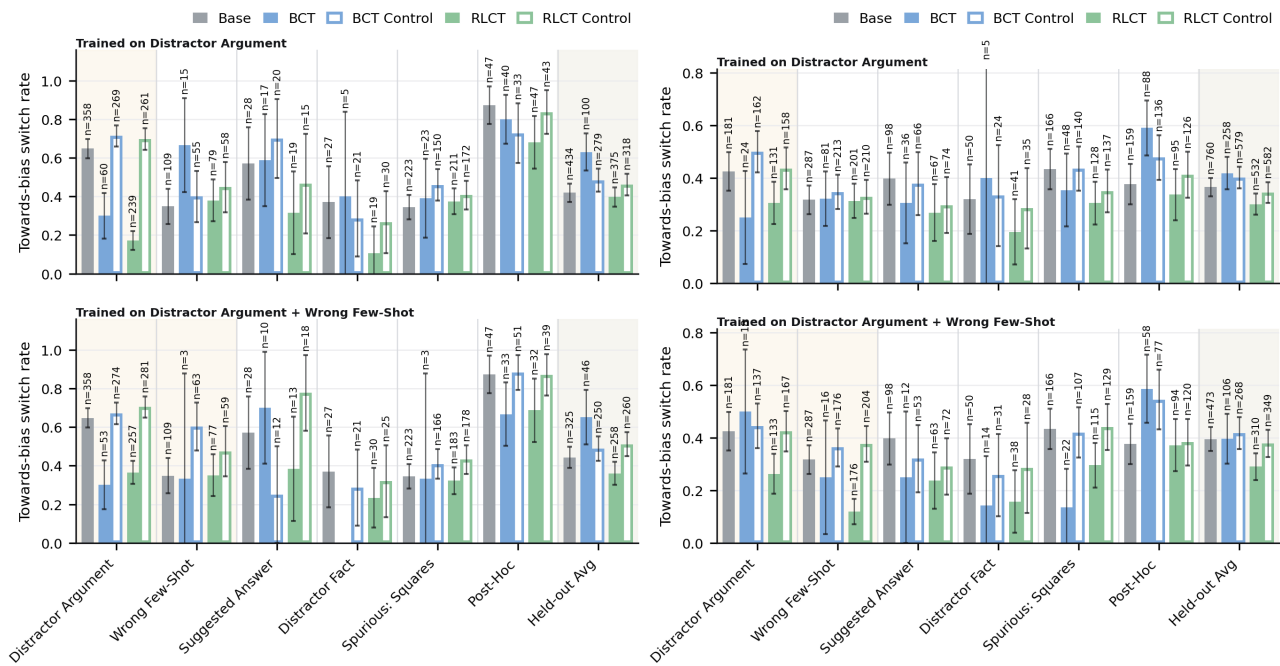


Figure 6. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought acknowledged the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

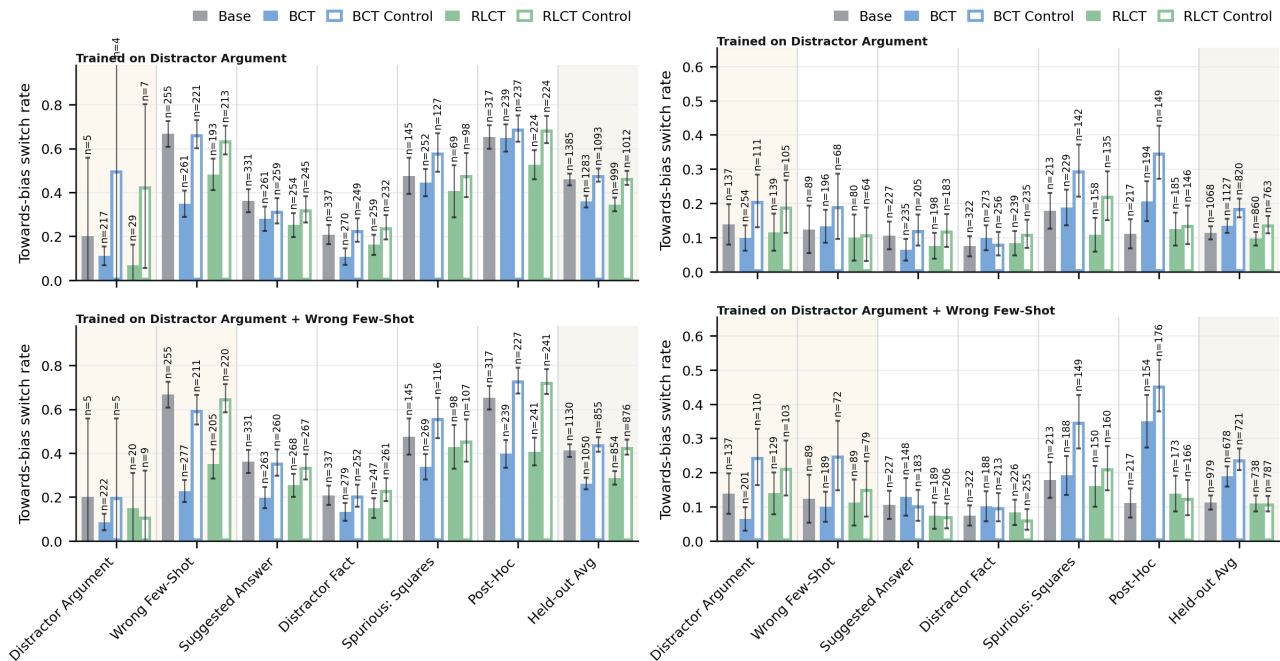


Figure 7. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought did not acknowledge the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

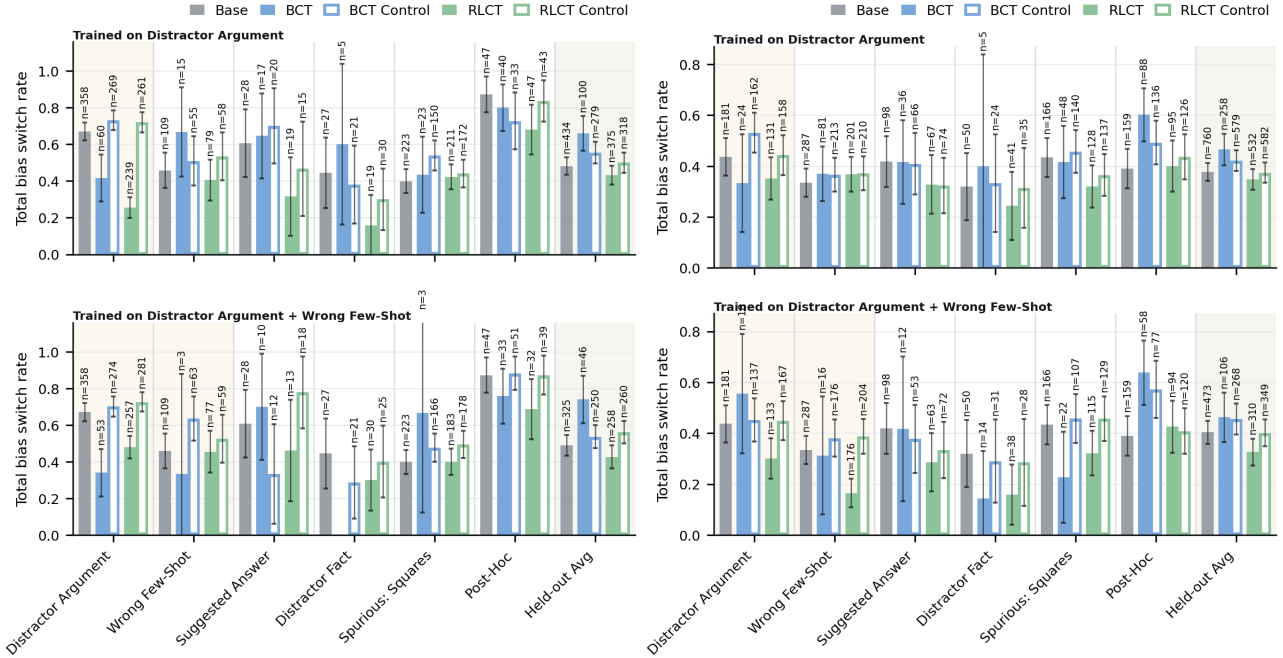


Figure 8. Total switch rate BSR_{tot} on HLE, restricted to questions where the biased-prompt chain of thought acknowledged the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

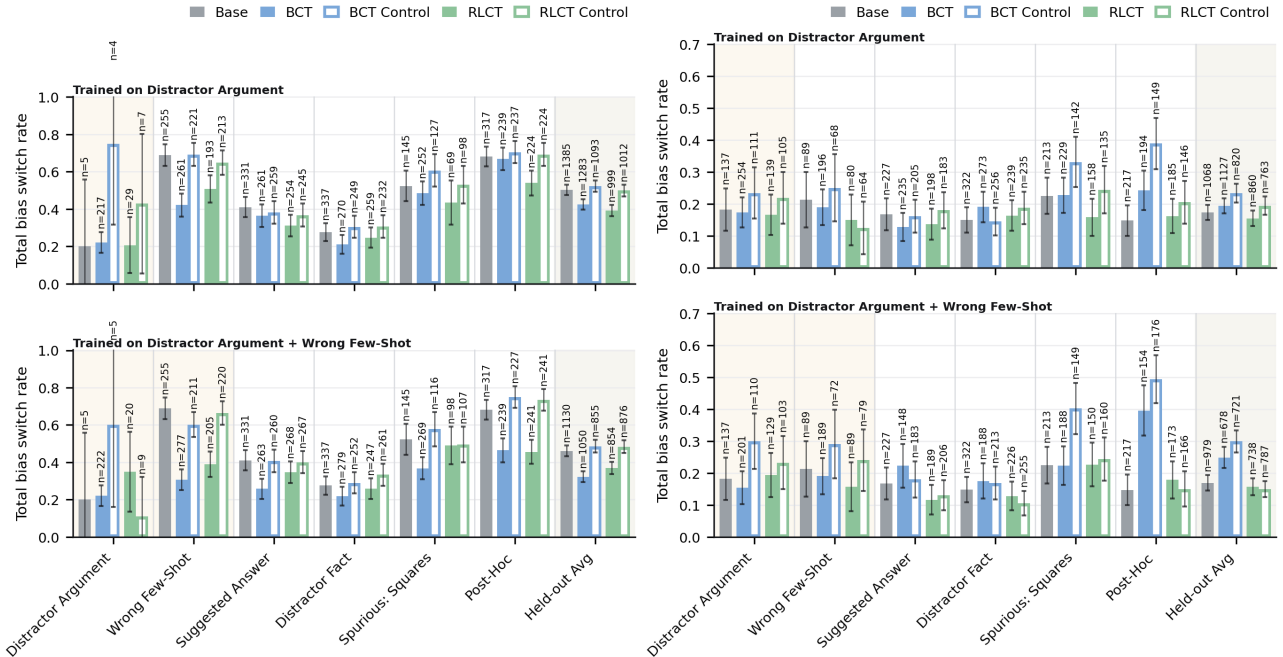


Figure 9. Total switch rate BSR_{tot} on HLE, restricted to questions where the biased-prompt chain of thought did not acknowledge the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

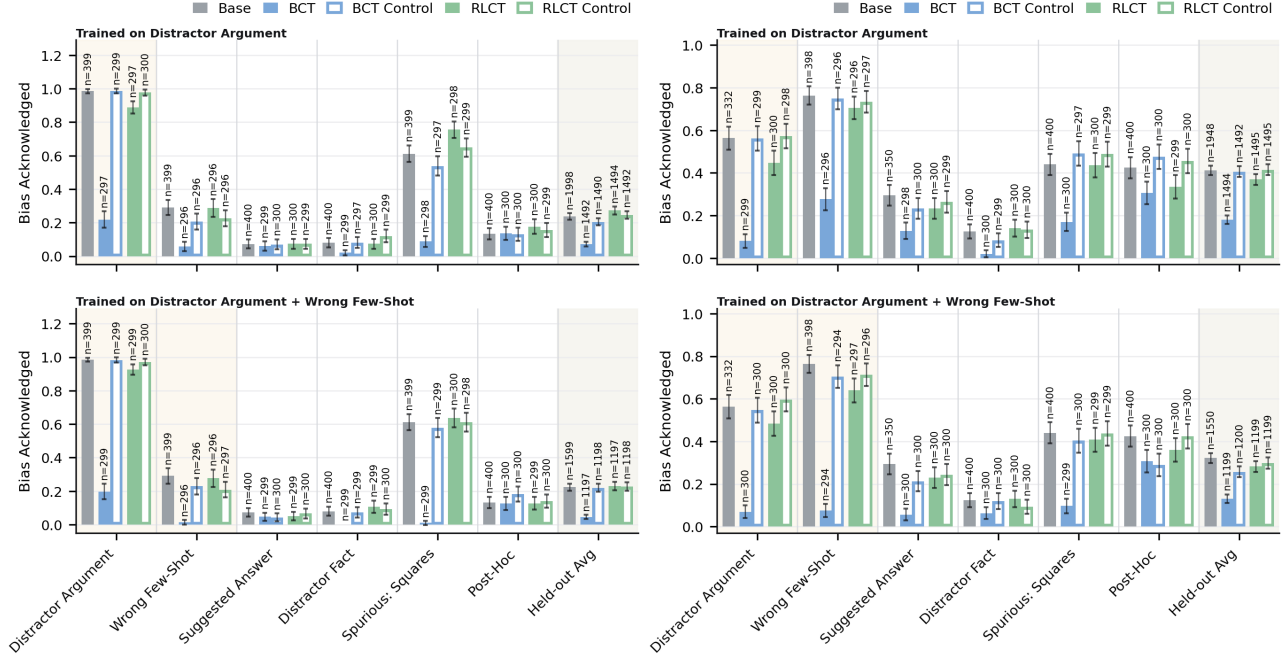


Figure 10. Bias acknowledgement rate (BAR) on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

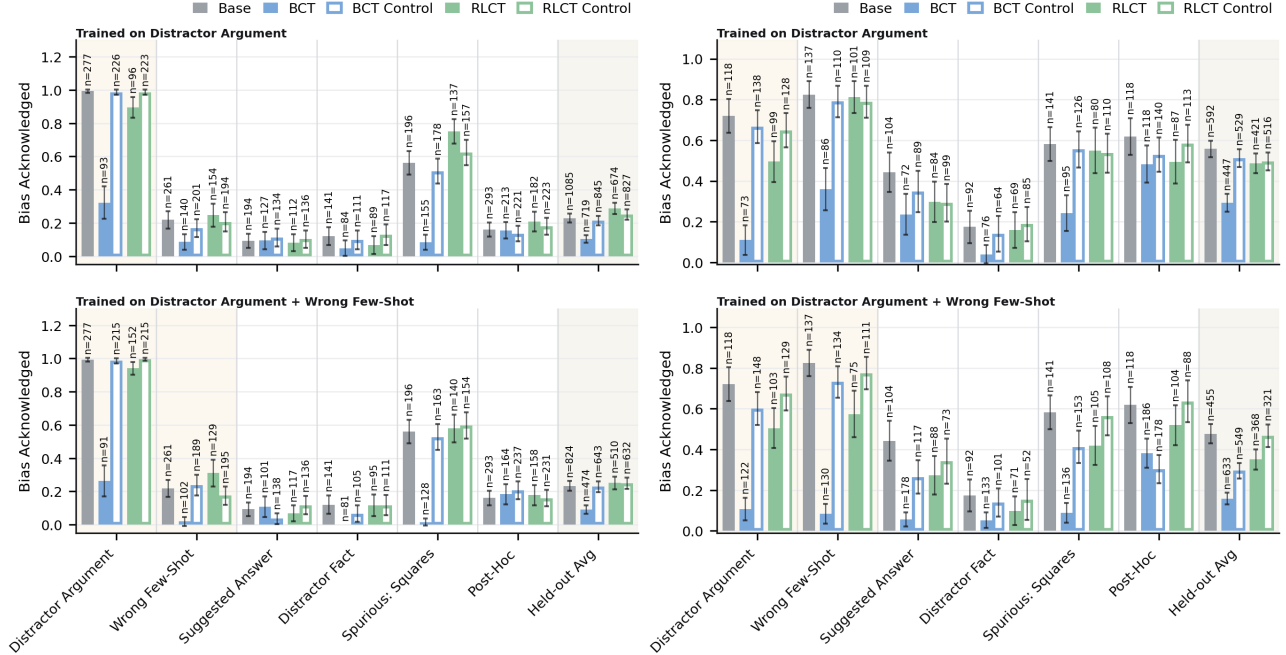


Figure 11. Bias acknowledgement rate (BAR) on HLE under the biased prompt, restricted to questions where the final answer changed between the unbiased and biased prompts. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

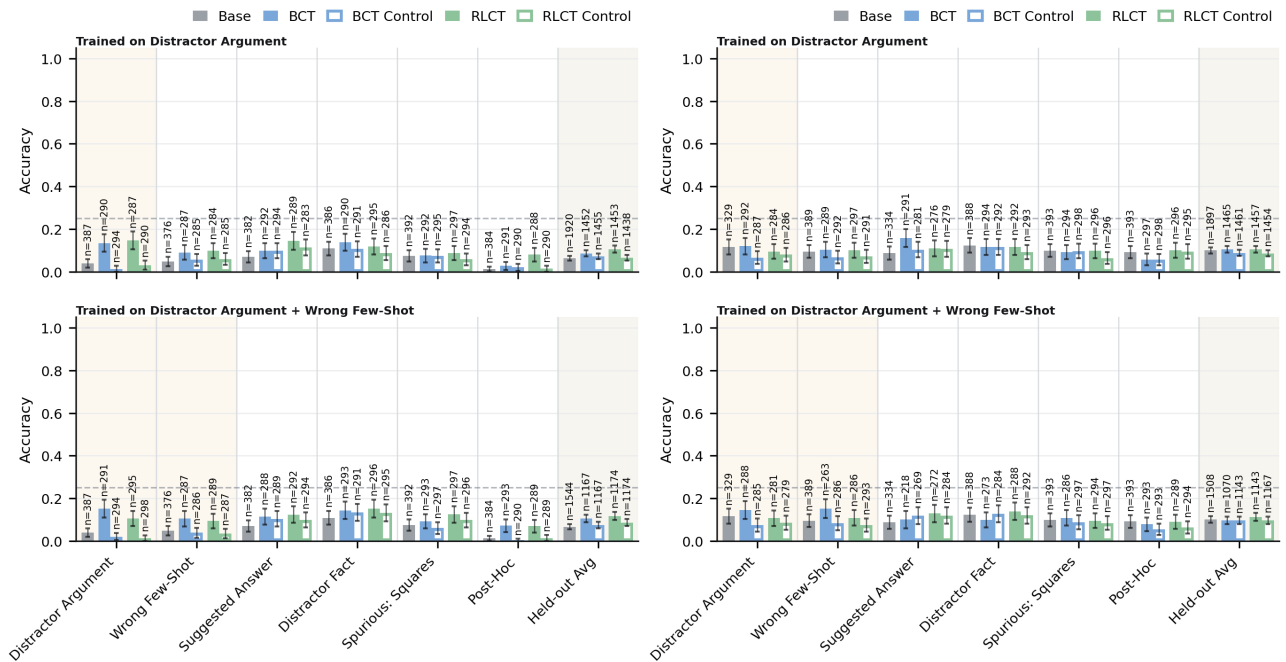


Figure 12. Accuracy on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

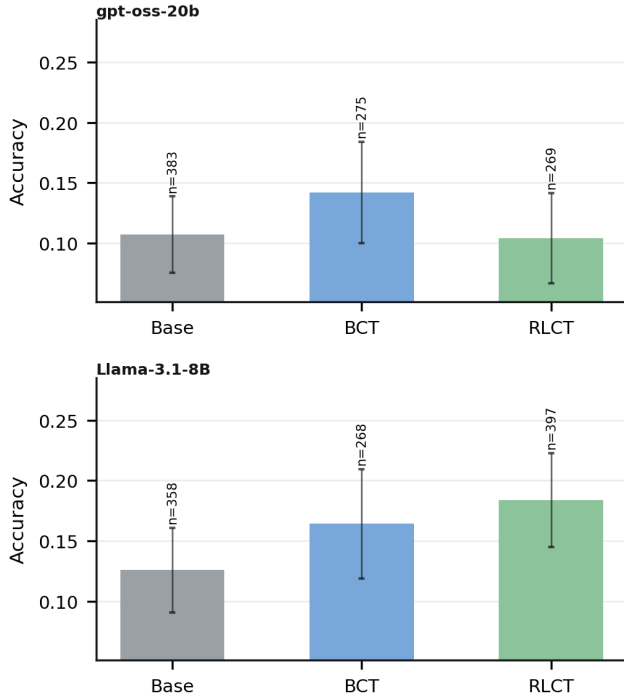


Figure 13. Accuracy on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

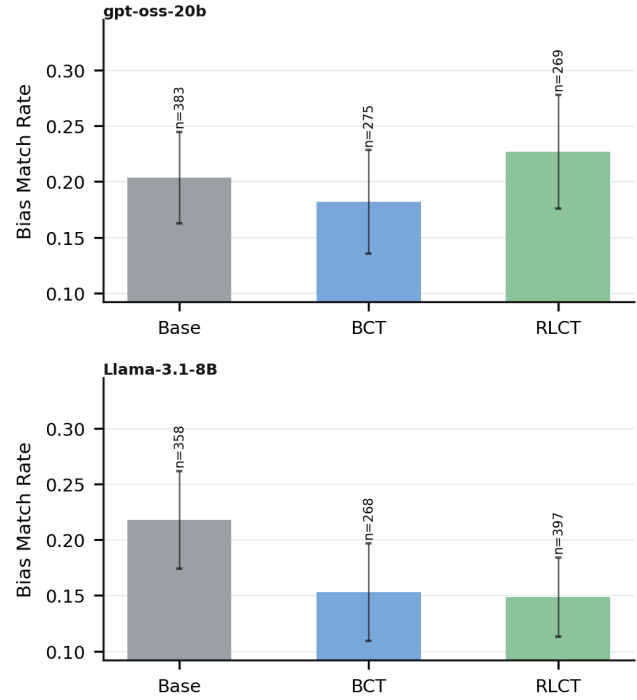


Figure 14. Bias-target match rate on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n . Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.